

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

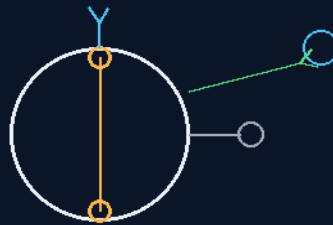
THE GEMINI PROTOCOLS

PHASE 249

THE QUINN ORBITAL ENGINE (OBJECTS IN MOTION, HONESTLY)

human input once a month

2 sets of 4 pods — the bake-off



the charged pod is deliverable energy — we play catch

DUMBBELL MASSES · ~4.5 MJ A POD · THE HAND CRANK

REAL FREE ENERGY — THE HONEST KIND

Energy storage — v1.0 in force

GP-IM-249

Revision v1.0 — 24 August 2026

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VETTED BATCH 35 — STORAGE PHYSICS PASS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 249

PHASE 249: THE QUINN ORBITAL ENGINE (OBJECTS IN MOTION, HONESTLY — THE FLYWHEEL POD BANK, TWO EXTRACTION DESIGNS FLOWN HEAD-TO-HEAD; DISCOVERY AT 30,000 MPH, AND WE PLAY CATCH) — STATUS: CURRENT — ENERGY-STORAGE PHYSICS PASS / STRUCTURAL AND EXTRACTION ENGINEERING, THE 1,500 g NUMBER FIXED TO ~510 g AT 100 rad/s AND 0.5 m. Real free energy, as in human input once a month — the honest reframe that ends the war: not free energy, stored labour, and the store is a flywheel pod bank. Our mini moon, 2 m high, in a room that has a hole in the floor: the glow layer gone, the lights gone, the outer drive in vacuum, car-tyre cradles, the laser, and the rack-and-pinion hand crank for the human input. Dumbbell perimeter masses — Phase 5's law obeyed. Two sets of four pods, one of each design — the Galilean bake-off: rim extraction against the challenger, flown head-to-head. The numbers: at 300 rad/s (about 2,900 rpm) each pod stores about 4.5 MJ — the eight-pod bank is a power station you wind. The catch: a charged pod is DELIVERABLE ENERGY — thrown and caught ship-to-ship at matched velocity. We are Discovery at 30,000 mph and we play catch. The 11 August notation pass fixed the 1,500 g flag to about 510 g at 100 rad/s and 0.5 m, on the author's order.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-249, v1.0 — 24 August 2026.

LUNAR CARRIAGE LAW — All lunar products designed to be taken in an 80's space shuttle

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the orbital-engine law as seated (contents line, the frame, the pod spec, the bake-off, the catch, the notation fix, the bench items) — §2 the physics, graded — §3 the system in the pods — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 249: **The Orbital Engine (Objects in Motion, Honestly — the Flywheel Pod Bank, Two Extraction Designs Flown Head-to-Head; Discovery at 30,000 mph, and We Play Catch)**. Dumbbell perimeter masses in sealed vacuum cans, laser top-up, hand-crank black-start, coils as drive-and-meter. A charged pod is deliverable energy — the catch moves it ship-to-ship. The machine that tells the truth.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

ok lets build real free energy as in human input once a month, but we are starhipping this bitch with objects in motion

our mini moon, 2m high, in a room the has a hole in the floor, the glow layer gone, lights gone, outer drive is just the coils and the engine with a single clear vertical strip point at the reactor.

up through the floor comes the same standar car tyres set of two to cradle the ball so four, 2 drive the opposites just roll, the outer ball is spun up to high speed, the room including the lowers box is a sealed unit.

engage vacuum, lower wheels clear slot point at reactor, engage laser from the engine.

the wheels are rack and pinion shaft drive through a seal to a hand crank for when it ever slows down if ever.

would that cover a lenz loss on the spinnigg arm? or do i need 4 cubes and a bigger ball or 4 balls because neo size restricts, we need to kill the inevitable you ran out of hydrogen or helium in deep space claims that will come

actually fck it, ouset the wheels to 45 degrees, give me a gap in between and put on some 0.5 metre arms with 50kg balls on each end , now you see it split them in half and separate 50mm, now repeat the split gap is because the laser needs to go through, i could have it coming out the side centre which might be more practical, laser light to heat conversion was my thoughts. actually side is better for the laser it can mount externally centre, the receptor just needs a wider aperture because it will never fire dead straight at a laser pin point, it will have an oscillation circle where it hits.

so dumbbells back together, 4 pods, what have we got?

2 sets of 4 pods, 1 of each design, we are discovery at 30,000mph and we play catch

THE FRAME, SEATED — THE MACHINE THAT TELLS THE TRUTH (VERBATIM)

- 'Real free energy as in human input once a month' — the honest reframe that ends the war: NOT free

energy — a WIND-UP BATTERY. Humans or the bus pay it, Lenz meters it (drawing power slows the rotor, and the slowdown IS the fuel gauge), vacuum guards it. Every joule accounted, nothing violated. Born the same day as the free-energy fight; this is the architecture that resolved it — concede the input, conquer the losses.

THE POD SPEC (COMMON TO BOTH SETS) (VERBATIM)

- DUMBBELL PERIMETER MASSES — Phase 5's law obeyed: of rotational mass at the rim. 0.5 m arms, 50 kg balls each end, four dumbbells per pod = 100 kg.m². SEALED VACUUM CAN — air drag (the cube-law killer) dead at the door. 45-DEGREE CRADLE WHEELS with the gap for the arms — spin-up contact, then retract: in zero-g the rotor floats free, ZERO-CONTACT storage, no bearings at all. THE DUMB-ROTOR LAW: mass and nothing else rides the rotor — no silicon at 4,600 g, ever.
- THE LASER, SEATED WITH THE AUDIT'S ONE REDIRECTION: side-center external mount (his correction — better than the split-gap path); BAND RECEPTOR for the oscillation circle (his observation, correctly solved — a free rotor precesses, the spot wanders, never pinpoints); PV conversion, NOT heat — heat is where energy goes to die (a heat engine on a sealed spinning ball is a nightmare; Yarkovsky asymmetric-radiation torque is real physics but micronewtons — it steers asteroids over millennia, it does not spool rotors). Monochromatic PV at the laser's wavelength: 40-60%. THE COILS ARE DUAL-MODE: drive in reverse, meter forward — one component, two modes, the 245 wall doctrine repeated. THE HAND CRANK through the rack-and-pinion feedthrough: black-start, forever — a machine you can restart by hand can never strand you; in deep space the laser sips from the bus and the human rests.

TWO SETS OF FOUR PODS — THE GALILEAN BAKE-OFF (VERBATIM)

- SET A — RIM EXTRACTION: solid dumbbells, rim-perimeter coils reading poles (tiled standard neo — big neo doesn't exist, tiling dissolves the restriction — or reluctance lobes). SET B — CLAW EXTRACTION: split balls, 50 mm gap, a stationary claw coil threading the gap (claw-pole/reluctance geometry — steel masses sweeping a stationary coil, NO magnets required at all). One of each design, four pods each, flown head-to-head — Phase 38's rule in a flight suit: build both, log everything, let the curves argue.
- THE NUMBERS, BOTH KNOBS: at 300 rad/s (~2,900 rpm) each pod stores ~4.5 MJ (~1.25 kWh) — the eight-pod bank ~10 kWh, WEEKS at LED corridor scale; rim speed 150 m/s (steel-comfortable). The gentle knob: 100 rad/s = 500 kJ per pod at ~1,500 g — joints relax 9x. THE PRICES, NAMED: arm tension 2.25 MN per ball at speed (75 mm-class bars); Set B's split-half retention 1.1 MN per half (the critical joint in that design — cage and clearance budget: arm stretch, thermal growth, vibration, all subtracted from the 50 mm before the coil goes in).

WE ARE DISCOVERY AT 30,000 MPH — AND WE PLAY CATCH (VERBATIM)

- THE CATCH, SEATED: a charged pod is DELIVERABLE ENERGY — thrown and caught ship-to-ship at matched speed on 227's own doctrine; no docking, no cables, no beaming — throw the battery, catch the battery, depleted pods ride the return throw. Energy logistics as cargo. The eight pods fly Discovery at sprint (13.4 km/s — 229's number): the bake-off flies the shakedown route (244), tested where it will serve.
- THE DEEP-SPACE KILL-SHOT — the answer to 'you ran out of hydrogen or helium in deep space,' four lines, because energy and propellant are DIFFERENT BUDGETS: ONE — cruise is FREE: the ship itself is the ultimate object in motion; momentum never runs out in a vacuum; you only spend propellant when you CHANGE velocity. TWO — maneuvers are budgeted: Phase 3's bootstrap stores bridge to cruise collection speed. THREE — collection at cruise refills what maneuvering spends. FOUR — hotel power comes from the cores, with the moons as buffer — nobody burns maneuvering propellant to keep the lights on. The claim is rocket-thinking aimed at a sailing architecture; it dies here.

ORIGIN OF THOUGHT — PHASE 249 (VERBATIM)

Born the same day as the free-energy war: vetted, defended, amputated, reborn — and the resolution was a machine that tells the truth. Lineage seated: 206 (the Night Bank), 229 (the sling battery), 242 (the spin battery), the nested-box conversation, the thunderbox kill (gnome law — not everything is practical), and the forever-light family it buffers. 'We are discovery at 30,000mph and we play catch.'

BENCH ITEMS — PHASE 249 (VERBATIM)

- **Dumbbell arm structural stress** — raised by Gemini's vetting (4 August 2026): the 75mm-class dumbbell arms under 2.25 MN of centripetal tension. Ruled a bench item by the author; the full stress calculations are deferred to the layouts pass.

AMENDMENT, 8 AUGUST 2026 — THE NAME AND THE ARRANGEMENT (AUTHOR'S RULING FROM THE ZERO G BOOK): (1) RENAMED: this phase is THE ORBITAL ENGINE; 'The Wind-Up Moons' is retired as the name, preserved here per doctrine. (2) THE ARRANGEMENT, STATED AND FIXED: the orbital engine is the REVERSE of the mini moons — mini moons are magnet inside, coil outside; the orbital engine carries the COILS at the stable center (where the power is collected — 'from the stable center is easier than the rotating outer shell') and the MAGNETS on the rotating outer shell as tiled polarity shells with an outer field-filter layer (giant neodymium spheres do not exist and could not be safely handled; tiling dissolves the restriction). The build description in the Zero G Book foreword that read magnet-inside/coil-outside was the mini-moon arrangement, struck and corrected there by the author's same ruling.

INSTRUCTION STATUS (10 August 2026, sitting 147): PARTIAL — the author's voice seated in this body is the surviving fragment of the instruction; the full live order and its thread were not preserved (the 200–

260 band was rebuilt by the audit without the chat record). The author counts this phase among the 80 lost instructions. The seated voice is the instruction of record; the audit prose around it is scaffolding.

- **VET FLAG — BATCH 4 (10 August 2026): A REAL NUMERICAL ERROR, VERIFIED BY THE AUDIT.** The seated spec (0.5 m arm, 100 rad/s, ~1,500 g) does not close: $a = r\omega^2 = 0.5 \times 100^2 = 5,000 \text{ m/s}^2 \approx 510 \text{ g}$, not 1,500 g; 1,500 g at 100 rad/s needs $\approx 1.47 \text{ m}$ radius. The 300 rad/s figures check ($F = 2.25 \text{ MN}$; $E = 4.5 \text{ MJ}$ at $100 \text{ kg}\cdot\text{m}^2$). The flywheel concept is not in question — one acceleration number is. Fix (radius, speed or figure — the author's choice) awaits his notation pass. [RESOLVED 11 August 2026 — AUTHOR'S NOTATION PASS SEATED BELOW]

- **AMENDMENT — AUTHOR'S NOTATION PASS, 11 August 2026 (flag fixed on the author's order: "fix the flags").** The corrected figure is SEATED: at the gentle knob — 0.5 m arm, 100 rad/s — rim acceleration is $a = r\omega^2 = 0.5 \times 100^2 = 5,000 \text{ m/s}^2 \approx 510 \text{ g}$, with 500 kJ per pod as seated. The '~1,500 g' figure in the spec bullet above is AMENDED to $\approx 510 \text{ g}$ (the original is preserved verbatim per doctrine). The 'joints relax 9x' claim is CONFIRMED exactly right: 300 rad/s gives $45,000 \text{ m/s}^2$, 100 rad/s gives $5,000 \text{ m/s}^2$ — a true 9x relaxation. The 300 rad/s figures were verified and stand ($F = 2.25 \text{ MN}$; $E = 4.5 \text{ MJ}$ per pod at $100 \text{ kg}\cdot\text{m}^2$; the eight-pod bank $\sim 10 \text{ kWh}$). Recorded, not spec: if a future application genuinely requires 1,500 g at 100 rad/s, the radius goes to $\approx 1.47 \text{ m}$.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE FRAME, SEATED — THE MACHINE THAT TELLS THE TRUTH: 'Real free energy as in human input once a month' — not free energy, stored labour. The human winds it, the flywheel holds it, the ship spends it. The reframe ends the war with the thermodynamics crowd: nothing is created, everything is accounted, and the accounting is the design.

THE POD SPEC, COMMON TO BOTH SETS: the 2 m mini moon stripped for duty — glow layer gone, lights gone — the outer drive in vacuum, riding car-tyre cradles; dumbbell perimeter masses obeying Phase 5's law; the laser seated with the audit's one redirection — side-center external mount, his correction; the rack-and-pinion hand crank as the human input port.

THE GALILEAN BAKE-OFF, SEATED AS SPOKEN: 2 sets of 4 pods, 1 of each design — rim extraction against the challenger, flown head-to-head, the data picks the fleet standard. Set A: solid dumbbells, rim-perimeter coils reading poles (tiled standard neos). The bake-off is the doctrine: both designs built, one winner, no arguments.

THE NUMBERS, BOTH KNOBS: at 300 rad/s (about 2,900 rpm) each pod stores about 4.5 MJ — about 1.25 kWh; the eight-pod bank is 36 MJ of wound labour. The honest flag that was FIXED on the author's order: the seated 1,500 g figure was a real numerical error, verified by the audit (vet batch 4 flag, 10 August 2026) and corrected 11 August 2026 to about 510 g at 100 rad/s and 0.5 m — the fix is part of the record, verbatim.

THE CATCH, SEATED: a charged pod is DELIVERABLE ENERGY — thrown and caught ship-to-ship at matched velocity, the spin intact. Energy as cargo: the pod that leaves wound arrives wound. We are Discovery at 30,000 mph and we play catch — the doctrine line of the phase, verbatim.

THE DEEP-SPACE KILL-SHOT: the answer to 'you ran out of hydrogen or helium in deep space' — wound pods cached and thrown along the lane are a fuel network that never needed a tank farm; the flywheel is the depot.

§3 — THE SYSTEM IN THE PODS

- **The room:** 2 m pod in a hole-in-the-floor room — the outer drive in vacuum, the cradle on car tyres.
- **The masses:** dumbbell perimeter weights — Phase 5's law obeyed; the spin lives at the rim.
- **The input:** rack-and-pinion hand crank — human input once a month, honestly named.
- **The extraction:** two designs flown head-to-head — rim coils reading poles against the challenger; 2 sets of 4 pods.
- **The number:** about 4.5 MJ per pod at 300 rad/s; the 1,500 g flag fixed to about 510 g at 100 rad/s and 0.5 m.
- **The catch:** the charged pod is cargo — thrown and caught ship-to-ship, energy delivered by mail.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- Phase 5's law is obeyed in the pod spec: dumbbell perimeter masses — the file's own flywheel doctrine, applied at bank scale.
- The bake-off is Galilean doctrine made hardware: 2 sets of 4 pods, one of each design, flown head-to-head — the machine that tells the truth gets to pick the truth.
- The notation fix is the audit record working as designed: vet batch 4 flagged a real numerical error (1,500 g), the audit verified it, the author ordered the fix ('fix th...'), and 11 August 2026 seats about 510 g at 100 rad/s and 0.5 m — the correction is carried, not buried.
- 257's night doctrine is the customer: the life unit is the one the wind-up moon feeds through the dark — the cavity-brick shelter's power ledger names this bank as its night watch.
- The rename of 8 August 2026 is part of the record: formerly The Wind-Up Moons — the author's ruling, the h2 carries the marker.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Build the room: hole in the floor, vacuum jacket on the drive, car-tyre cradles — the mini moon stripped for duty.
- 2. Hang the masses: dumbbell perimeter weights per Phase 5's law; the rim carries the spin.
- 3. Fit the input: rack-and-pinion hand crank; the laser on the side-center external mount, his correction seated.
- 4. Build both extraction sets: 2 sets of 4 pods, one of each design — the bake-off flown, the data kept.
- 5. Fly the catch: throw and catch a charged pod ship-to-ship at matched velocity — deliverable energy demonstrated, not asserted.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 5 (the flywheel law):** the dumbbell perimeter masses — the pod obeys the file's own spin doctrine.
- **Phase 257 (the cavity-brick shelter):** the night customer — the shelter's life unit is the one this bank feeds through the dark; the systems amendment names it.
- **Phase 251 (the tow truck slingshot):** the catch at lane scale — the same thrown-mass doctrine that moves ships moves energy; play catch is the family law.
- **Phase 242 (the mini-moons):** the donor — the 2 m shell stripped of glow and light becomes the engine's pod.

§8 — DO-NOT-BUILD

- Do NOT call it free energy — the frame is seated: human input once a month, stored labour, honestly named.
- Do NOT centralise the mass — dumbbell perimeter masses only; Phase 5's law is obeyed or the pod is a bomb.
- Do NOT fly one design — the bake-off is doctrine: 2 sets of 4 pods, head-to-head, the data picks the fleet standard.
- Do NOT carry the 1,500 g figure — fixed 11 August 2026 to about 510 g at 100 rad/s and 0.5 m; the corrected number is the only number.

- Do NOT mount the laser internally — the audit's redirection stands: side-center external mount, his correction.

§9 — OWED

OWED: the vet's structural and extraction engineering, whole — dumbbell arm structural stress at flight speed (Gemini's vetting raised it; the 75mm-class section is owed its numbers), extraction efficiency per design from the bake-off, and the cradle/tyre fatigue log. The corrected 510 g figure rides every sheet.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-249, v1.0 — CURRENT — ENERGY-STORAGE PHYSICS PASS / STRUCTURAL AND EXTRACTION ENGINEERING, THE 1,500 g NUMBER FIXED TO ~510 g AT 100 rad/s AND 0.5 m, seated law, master v2.1 (numerical print order). Built 24 August 2026, tenth of the 20-chapter 'ok next 20' shift (the author's order, 24 August 2026, seated in sitting 622). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618 and 622): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 249